

## CHAPTER TWO: FOUNDATIONAL CONCEPTS IN MIS

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## CHAPTER TWO: FOUNDATIONAL CONCEPTS IN MIS

### 2.1 INTRODUCTION

Information systems and technologies are a vital component of successful businesses and organizations. They thus constitute an essential field of study in business administration and management. Since you probably intend to be a manager, entrepreneur, or business professional, it is just as important to have a basic understanding of information systems as it is to understand any other functional area in business. Information technologies, including Internet-based information systems, are playing a vital and expanding role in business. Information technology can help all kinds of businesses improve the efficiency and effectiveness of their business processes, managerial decision making, and workgroup collaboration, thus strengthening their competitive positions in a rapidly changing marketplace.

An **information system** (IS) can be any organized combination of people, hardware, software, communications networks, data resources, and policies and procedures that stores, retrieves, transforms, and disseminates information in an organization. People rely on modern information systems to communicate with each other using a variety of physical devices (*hardware*), information processing instructions and procedures (*software*), communications channels (*networks*), and stored data (*data resources*).

### **The new role of IS in organizations**

Managers can't ignore technology anymore and pass it off to secretaries or clerical workers or the Information Technology department. Information Systems are critical to the success of organization at all managerial levels. There is a growing interdependence between business strategy, rules, and procedures on the one hand and information systems software, hardware, databases and telecommunications on the other. A change in any of these components often requires changes in other components. This relationship becomes critical when management

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plans for the future. What a business would like to do in five years is often dependent on what its systems will be able to do. Increasing market share, becoming the high quality or low-cost producer, developing new products, and increasing employee productivity depend more on the kinds and quality of information systems in the organization. A second change in the relationship of information systems and organizations results from the growing complexity and scope of systems projects and applications. Building systems today involves a much larger part of the organization than it did in the past. Whereas early systems produced largely technical changes that affected few people

### **How Information Technology has changed the Management Process**

Times have changed and so have the methods by which managers make decisions. Information technology has helped speed the change in methods.

#### ***Traditional and Contemporary Management***

Technology has enabled companies to flatten their hierarchies. The last few years has seen an exodus of middle managers. Companies simply didn't need the extra layers because of technological advances that allow lower levels of employees to communicate and collaborate easier and faster than ever before. Managers in these newly flattened organizations are now responsible for making sure employees know the environmental influences on the organization, know the goals of the organization, and adjust the organization to meet the new influences. Managers then free their employees to meet not only the organization's goals, but also their personal goals. Information systems can help managers and employees work more efficiently and effectively in this new environment by increasing the amount of information available to all employees. Communications are faster and more widespread with new technologies that enable employees and managers to collaborate more closely and work better in teams. New information systems also enable virtual organizations and geographically dispersed teams and groups to work together to meet personal and organizational goals.

#### ***Implications for System Design***

The decision-making process is much different in today's organization than it was just a few years ago. The danger of building a system to accommodate today's process is that it will not take these changes into account. Understanding how people and organizations make decisions will help build a system that can accommodate the organization and the employees. Information systems should be created not only to help managers and employees make decisions, but also help them better communicate between all levels and units of the

organization. Remember, decisions are not made in isolation. More important, decisions affect a wide range of people, and the system should accommodate this fact. The real danger in using information systems to help make decisions is that the decision-making process will be based on the wrong information. Because managers may assume that the situation is similar to one they experienced before, they may not be as careful as they would be if it were an entirely new situation. For instance, management may decide that the new packaging materials are as good as the old ones because they are the same color. Therefore managers won't be as careful in studying all the data, all the possible outcomes and the alternatives when making the decision to change suppliers. They make the decision based on the first available alternative that moves them toward their ultimate goal. They find out too late that the packaging materials are not as good as the old ones and they end up with more damaged goods and irate customers.

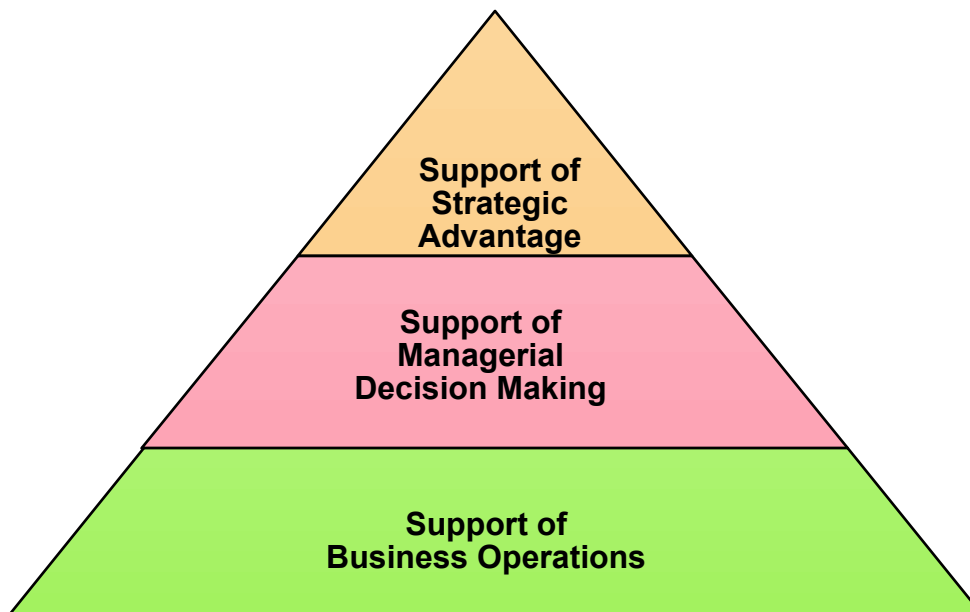
Information systems have these features:

- They are flexible and provide many options for handling data and evaluating information.
- They are capable of supporting a variety of styles, skills, and knowledge.
- They are powerful in the sense of having multiple analytical and intuitive models for the evaluation of data and the ability to keep track of many alternatives and consequences.
- They reflect the bureaucratic and political requirements of systems.
- They reflect an appreciation of the limits of organizational change and an awareness of what information systems can and cannot do.

### **The Fundamental Roles of IS in Business**

There are three fundamental reasons for all business applications of information technology. They are found in the three vital roles that information systems can perform for a business enterprise. These are:

- A. Support of its business processes and operations,
- B. Support of decision making by its employees and managers, and
- C. Support of its strategies for competitive advantage.



**Fig.2.1: Major Roles of Information Systems**

**Support Business Processes:** As a consumer, you regularly encounter information systems that support the business processes and operations at the many retail stores where you shop. For example, most retail stores now use computer-based information systems to help their employees record customer purchases, keep track of inventory, pay employees, buy new merchandise, and evaluate sales trends. Store operations would grind to a halt without the support of such information systems.

**Support Decision Making:** Information systems also help store managers and other business professionals make better decisions. For example, decisions on what lines of merchandise need to be added or discontinued, or on what kind of investment they require, are typically made after an analysis provided by computer-based information systems. This not only supports the decision making of store managers, buyers, and others, but also helps them look for ways to gain an advantage over other retailers in the competition for customers.

**Support Competitive Advantage:** Gaining a strategic advantage over competitors requires innovative application of information technologies. For example, store management might make a decision to install touch-screen kiosks in all of their stores, with links to their e-commerce website for online shopping. This might attract new customers and build customer loyalty because of the ease of shopping and buying merchandise provided by such information systems. Thus, strategic information systems can help provide products and services that give a business a comparative advantage over its competitors.

## 2.2. MANAGEMENT AND BUSINESS FUNCTIONS

Every aspect of management in the modern age relies heavily on information to thrive. Nothing moves without information and it is generally believed that information is power and that he who has it has power. It is an important resource needed to develop other resources. Changing circumstances and environments have necessitated the need for the proper dissemination of information at various levels of management. The development and use of information management systems (MIS) is a modern phenomenon concerned with the use of appropriate information that will lead to better planning, better decision making and better results.

### 2.2.1. Management Function



### 2.2.2. Business Functions

The primary function of MIS is to help a manager take an action, answer a question or ask the right question. Organizations differ considerably in their activities. However, their strategies and the way in which they seek to add value we can identify some functions which they usually have in common. There may be vary many functions for an organization, like Production/Operations, Marketing, Finance, Human Resource Management, Logistics, Information Technology & Communications, etc. Some of the most common business functions in Organizations are explained below:

a. **Production/ Operations:** this involves the actual production and delivery of the product or service. In the primary sector this may mean growing the product (e.g. farming) or extracting it (e.g. oil); in the secondary sector this involves activities such as assembly, manufacture and construction and in the tertiary sector this involves providing a service such as tourism, education and insurance. Operational decisions include deciding where to produce, how to produce, what volume and range of products to produce and what quality and cost targets to achieve. It also involves research and development into new products and processes.

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b. **Marketing:** the marketing activities of business begin with identifying customer needs. Having identified customers' requirements marketing activities aim to satisfy these needs through ensuring the firm provides the right products, at the right place and price and at the right time. The marketing mix involves the 4Ps: deciding on the price, the product itself, the promotion and the place.

c. **Finance:** Organizations need to raise finance to get started and to invest into new projects. Firms also need to set financial targets and allocate money within the business; this is known as budgeting. Budgets will be set for a given period in the future and then compared with the actual outcomes to examine why differences occurred; this is known as variance analysis. Organizations will also produce financial reports to their investors such as balance sheets and the profit and loss account.

d. **Human resource management (HRM):** All organizations rely on their employees and HRM refers to the way in which people are managed. HRM involves activities such as the recruitment and selection of staff, the training of people and the development and implementation of appropriate reward systems. The way that people are managed will influence whether they turn up for work, how productive they are and their openness to change.

**The Information System Function:** Successful management of information systems and technologies presents major challenges to business managers and professionals. Thus, the information systems function represents:

- A major functional area of business equally as important to business success as the functions of accounting, finance, operations management, marketing, and human resources management.
- An important contributor to operational efficiency, employee productivity and morale, and customer service and satisfaction.
- A major source of information and support needed to promote effective decision making by managers and business professionals.
- A vital ingredient in developing competitive products and services that give an organization a strategic advantage in the global marketplace.
- A dynamic, rewarding, and challenging career opportunity for millions of men and women.

- A key component of the resources, infrastructure, and capabilities of today's networked business enterprises.

It is often said that the use of information technology makes our work more effective, more efficient, or both. What do these terms mean?

**Effectiveness** defines the degree to which a goal is achieved. Thus, a system is more or less effective depending upon **(1)** how much of its goal it achieves, and **(2)** the degree to which it achieves better outcomes than other systems do.

**Efficiency** is determined by the relationship between resources expended and the benefits gained in achieving a goal. Expressed mathematically,

$$\text{Efficiency} = \frac{\text{Benefits}}{\text{Costs}}$$

Thus, one system is more efficient than another if its operating costs are lower for the same or better quality product, or if its product's quality is greater for the same or lower costs. The term productivity is commonly used as a synonym for efficiency. However, productivity specifically refers to the efficiency of human resources. Productivity improves when fewer workers are required to produce the same amount of output, or, alternately, when the same number of workers produce a larger output. The closer the result of an effort is to the ultimate goal, the more effective the effort. The fewer the resources spent on achieving a goal, the more efficient the effort.

### **2.3 DATA, INFORMATION, KNOWLEDGE AND WISDOM (DIKW PYRAMID)**

The DIKW Pyramid, also known variously as the "DIKW Hierarchy", "Wisdom Hierarchy", the "Knowledge Hierarchy", the "Information Hierarchy", and the "Knowledge Pyramid", refers loosely to a class of models for representing purported structural and/or functional relationships between data, information, knowledge, and wisdom. The presentation of the relationships among data, information, knowledge, and sometimes wisdom in a hierarchical arrangement has been part of the language of information science for many years. Although it is uncertain when and by whom those relationships were first presented, the ubiquity of the notion of a hierarchy is embedded in the use of the acronym DIKW as a shorthand representation for the data-to-information-to-knowledge-to-wisdom transformation.

**Data:** The word **data** is the plural of *datum*, though data commonly represents both singular and plural forms. Data is conceived of as symbols or signs, representing stimuli or signals, that are of no use until...in a usable (that is, relevant) form. Data consists of representations of events, people, resources, or conditions. The representations can be in a variety of forms, such as numbers, codes, text, graphs, or pictures. For example, a spacecraft launch or the sale of an automobile would generate a lot of data describing those events.

**Data:**

- A critical component of information systems.
- Information systems typically, collect store and process data.
- Business data is collected in a number of different ways.
- Primarily via the system input devices such as keyboards, a mouse, scanners, etc.
- Typically stored in the various associated computer system storage devices, databases, for example.
- Operational information systems collect and store data every time a transaction is carried out

**Information:** is useful facts extracted from data which is used for decision-making. Information is contained in descriptions, and is differentiated from data in that it is "useful". Information is defined as data that are endowed with meaning and purpose. Information is a result of processing data. It provides the recipient with some understanding, insight, conclusion, decision, confirmation, or recommendation. The information may be a report, an analysis, data organized in a meaningful output, a verbal response, a graph, picture, or video.

**Information:**

- 1) Their form is aggregated, manipulated, and organized;
- 2) Their content is analyzed and evaluated; and
- 3) They are placed in a proper context for a human user.

The issue of context is really at the heart of understanding the difference between information and data. Data can be thought of as context-independent: A list of numbers or names, by itself, does not provide any understanding of the context in which it was recorded. In fact, the same list could be recorded in a variety of contexts. In contrast, for data to become information, both the context of the data and the perspective of the person accessing the data become essential. The same data may be considered valuable information to one person and completely irrelevant to the next. Just think of data as potentially valuable to all and the value

of information as being valuable relative to its user. Information systems transform data into information. For example, data shown earlier transformed

- ✓ Customer ID: 5002100
- ✓ Price: 97.99
- ✓ Date:10-8-06

**Knowledge:** is a fluid mix of framed experience, values, contextual information, expert insight and grounded intuition that provides an environment and framework for evaluating and incorporating new experiences and information. It originates and is applied in the minds of knower's. In organizations it often becomes embedded not only in documents and repositories but also in organizational routines, processes, practices and norms.

**Wisdom:** is the ability to increase effectiveness. Wisdom adds value, which requires the mental function that we call judgment. The ethical and aesthetic values that this implies are inherent to the actor and are unique and personal. We can say that wisdom is the practical application of knowledge in to problem-solving. In other words, knowledge is information organized and processed to convey understanding, experience, accumulated learning, and expertise. It provides the basis for action. Knowledge may be procedural (how to do something), formal (general principles, concepts, and procedures), tacit (expertise from experience that is somewhat hidden), and meta knowledge (knowledge about where knowledge is to be found).

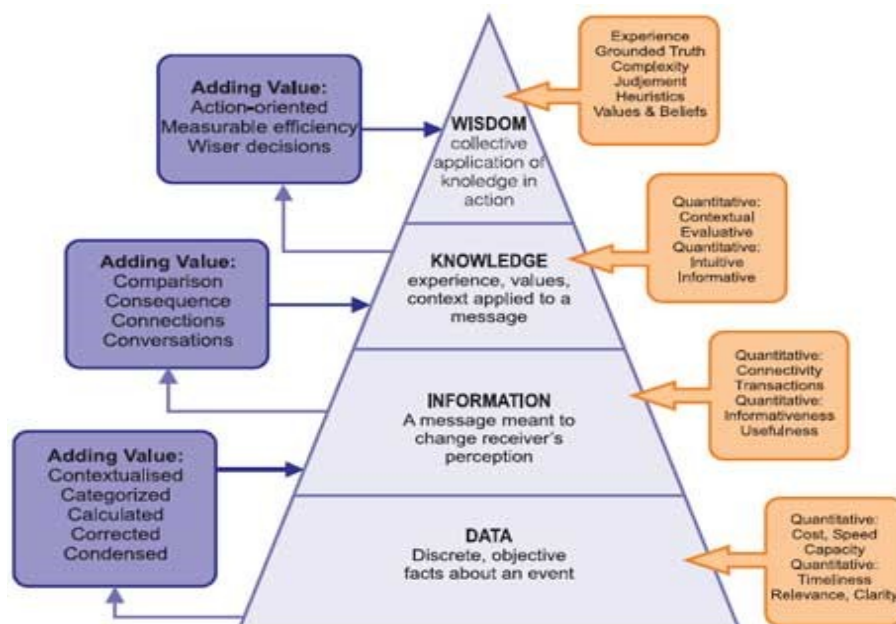


Figure 2.1 DIKW Pyramid

**Characteristics of Information:**

**I) Timeliness:** Timeliness means that information must reach the recipients within the prescribed timeframes. For effective decision-making, information must reach the decision-maker at the right time, i.e. recipients must get information when they need it. Delays destroys the value of information. The characteristic of timeliness, to be effective, should also include up-to-date, i.e. current information.

**II) Accuracy:** Information should be accurate. It means that information should be free from mistakes, errors & it must be clear. Accuracy also means that the information is free from bias. Wrong information given to management would result in wrong decisions. As managers decisions are based on the information supplied in MIS reports, all managers need accurate information.

**III) Relevance:** Information is said to be relevant if it answers especially for the recipient what, why, where, when, who and why? In other words, the MIS should serve reports to managers which is useful and the information helps them to make decisions.

**IV) Adequacy:** Adequacy means information must be sufficient in quantity, i.e. MIS must provide reports containing information which is required in the deciding processes of decision-making. The report should not give inadequate or for that matter, more than adequate information, which may create a difficult situation for the decision-maker. Whereas inadequacy of information leads to crises, information overload results in chaos.

**V) Completeness:** The information which is given to a manager must be complete and should meet all his/her needs. Incomplete information may result in wrong decisions and thus may prove costly to the organization.

**VI) Explicitness:** A report is said to be of good quality if it does not require further analysis by the recipients for decision making.

**VII) Impartiality:** Impartial information contains no bias and has been collected without any distorted view of the situation.

## **2.4 THE INFORMATION NEEDS AND SOURCES OF MANAGERS**

Businesses and other organizations need information for many purposes. Anyhow, we can summarize the five main uses of information in business firms as below:

**A. Planning:** To plan properly, a business needs to know what resources it has (e.g. cash, people, machinery and equipment, property, customers). It also needs information about the markets in which it operates and the actions of competitors. At the planning stage,

information is important as a key ingredient in decision-making. Hence, information regarding external environment and internal environment are essential at planning stage.

**B. Executing:** Executing is the stage where the plans are being implemented at organizations. Even during the phase of execution, managers require information about the operational environment to make sure that the process of implementation is taking place in the right path. Moreover, information is essential at this stage to assess the effects of execution.

**C. Reporting:** Information is essential to be used at reporting, especially reporting to the superiors within the organization and reporting to external bodies such as government authorities.

**D. Resource Appraisal:** At the stage of resource appraisal, especially employee appraisal, information about the resource performances are vital.

**E. Decision-making:** Information used for decision-making is often categorized into three types:

- i. **Strategic Information:** used to help plan the objectives of the business as a whole and to measure how well those objectives are being achieved.
- ii. **Tactical Information:** this is used to decide how the resources of the business should be employed.
- iii. **Operational Information:** this information is used to make sure that specific operational tasks are carried out as planned/intended (i.e., things are done properly).

When it comes to the **sources of information**, we can broadly classify various sources of information into *Primary Sources* and *Secondary Sources*. Although, neither of them is superior to each other, depending on decision-making situations importance varies on them.

**A. Primary Sources:** of information are the original materials that the decision-maker has gathered specifically for the purpose of solving the particular problem. This includes information collected through surveys, observations, questionnaires, experiments, opinion polls, etc.

**B. Secondary Sources:** of information are that information gathered by the decision-maker from already existing sources. They are not specifically gathered for his existing problem, but reused by the decision-maker from other sources such as magazines, news papers, reports, archives, databases, etc.

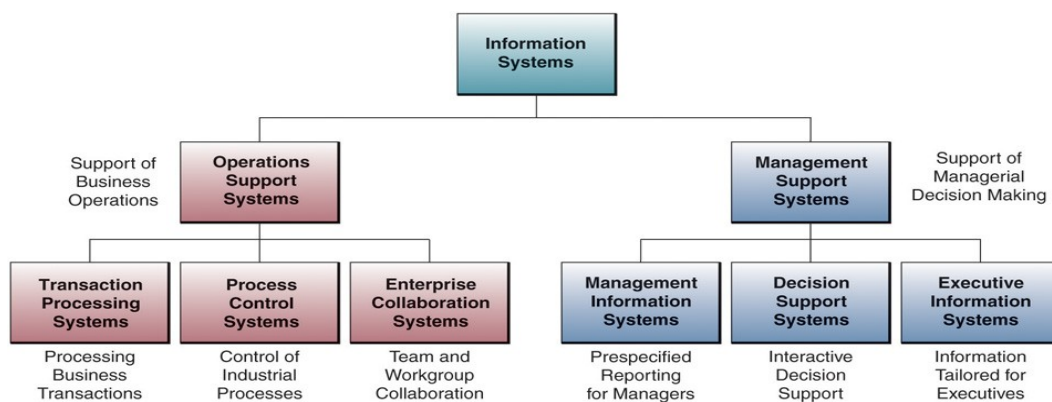
## **2.5 BUSINESS SYSTEMS (INFORMATION SYSTEM TYPES)**

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A business system, also known as a business information system, is a structure put in place within an organization with the purpose of improving effectiveness and efficiency. There can many kinds of business systems within an organization, with some touching on only one specific area of the business. Business systems help organizations function on a daily basis and in the long term. Some of the most common *Business Systems* are as below:

### **1.12 TYPES OF INFORMATION SYSTEM/BUSINESS SYSTEMS**

A typical business organization has systems supporting processes for each of the major business functions systems for sales and marketing, manufacturing and production, finance and accounting, and human resources. A business firm has systems to support different groups or levels of management:



#### **1. Transaction Processing System (TPS)**

Operational managers need systems that keep track of the elementary activities and transactions of the organization, such as sales, receipts, cash deposits, payroll, credit decisions, and the flow of materials in a factory. Transaction processing systems (TPS) provide this kind of information. A transaction processing system is a computerized system that performs and records the daily routine transactions necessary to conduct business, such as sales order entry, hotel reservations, payroll, employee record keeping, and shipping. Computerized systems that perform and record the *daily routine transactions* necessary to conduct the business; they serve the organization's *operational level*.

#### ■ Examples:

- ATM Machine System Banking Transactions
- Cash Register System Point of Sale Transactions
- Accounting System Checking Account Transactions
- Even Pay-per-view or On Demand is a TPS

## 2. Process Control Systems (PCS)

Process Control systems (PCS) provide this kind of information which provides the suitable information about the process and supports production. It continuously monitors the process in manufacturing.

- Supports Operations, Monitor and Control Industrial/Manufacturing Process
- Examples: Petroleum Refining, Power Generation, Automobile Manufacturing

## 3. Enterprise Collaboration Systems (ECS)

An Enterprise collaboration system (ECS) provides the suitable information among the participants of the organisation. It continuously expects the teamwork and collaboration by proper communication through various medias available..

- Teamwork, communication, and collaboration
- Examples: E-mail, Chat, Video Conferencing, Calendaring, Journaling

## 4. Management Information System (MIS)

Middle management needs systems to help with monitoring, controlling, decision-making, and administrative activities. Management information system is the study of information systems in business and management. The term management information systems (MIS) also designates a specific category of information systems serving middle management. MIS provide middle managers with reports on the organization's current performance. This information is used to monitor and control the business and predict future performance. MIS summarize and report on the company's basic operations using data supplied by transaction processing systems. The basic transaction data from TPS are compressed and usually presented in reports that are produced on a regular schedule. Today, many of these reports are delivered online. MIS serve managers primarily interested in weekly, monthly, and yearly results. These systems typically provide answers to routine questions that have been specified in advance and have a predefined procedure for answering them.

## 5. Decision-Support System (DSS)

The decision-support systems (DSS) support more non-routine decision making. They focus on problems that are unique and rapidly changing, for which the procedure for arriving at a solution may not be fully predefined in advance. They try to answer questions such as these: What would be the impact on production schedules if we were to double sales in the month of December? What would happen to our return on investment if a factory schedule were delayed for six months?

Although DSS use internal information from TPS and MIS, they often bring in information from external sources, such as current stock prices or product prices of competitors. These systems use a variety of models to analyze the data and are designed so that users can work with them directly. All of the management systems we have just described are systems for business intelligence. **Business intelligence** is a contemporary term for data and software tools for organizing, analyzing, and providing access to data to help managers and other enterprise users make more informed decisions.

## 6. Executive Support System (ESS)

Executive support systems (ESS) help senior management make these decisions. They address non-routine decisions requiring judgment, evaluation, and insight because there is no agreed-on procedure for arriving at a solution. ESS present graphs and data from many sources through an interface that is easy for senior managers to use. Often the information is delivered to senior executives through a portal, which uses a Web interface to present integrated personalized business content.

ESS are designed to incorporate data about external events, such as new tax laws or competitors, but they also draw summarized information from internal MIS and DSS. They filter, compress, and track critical data, displaying the data of greatest importance to senior managers. Increasingly, such systems include business intelligence analytics for analyzing trends, forecasting, and drilling down to data at greater levels of detail.

- Supports high-level strategic management
- Provides critical info from other systems (MIS and DSS).
- **Portal Concept:** one place with links to all information
- EIS s integrate **external information** such as economic developments and news about related markets and competitors. Helps strategic decision making, not necessarily tactical.
- Tactical doing things the right way right
- Strategic doing the right things

## 7. Knowledge Management System (KMS)

Knowledge Management Systems some firms perform better than others because they have better knowledge about how to create, produce, and deliver products and services. This firm knowledge is difficult to imitate, unique, and can be leveraged into long-term strategic benefits. Knowledge management systems (KMS) enable organizations to better manage processes for capturing and applying knowledge and expertise. These systems collect all

relevant knowledge and experience in the firm, and make it available wherever and whenever it is needed to improve business processes and management decisions. They also link the firm to external sources of knowledge. Information systems that aid knowledge workers in the *creation and integration of new knowledge* in the organization. Computer systems, such as word processing, electronic mail systems, and scheduling systems, that are designed to *increase the productivity of data workers* in the office .

**2.6.1. E-Business or Electronic Business:** is a term used to describe businesses run on the Internet, or utilizing Internet technologies to improve the productivity or profitability of a business. In a more general sense, the term may be used to describe any form of electronic business - that is to say, any business which utilizes a computer. This usage is somewhat archaic, however, and in most contexts e-business refers exclusively to Internet businesses. The Internet and related technologies and applications have changed the way businesses are operated and people work, and how information systems support business processes, decision making, and competitive advantage. Thus, many businesses today are using Internet technologies to Web-enable business processes and to create innovative e-business applications. E-business is defined as the use of Internet technologies to work and empower business processes, electronic commerce, and enterprise collaboration within a company and with its customers, suppliers, and other business stakeholders. In essence, e-business can be more generally considered *an online exchange of value*. Any online exchange of information, money, resources, services, or any combination thereof, falls under the e-business umbrella. The Internet and Internet-like networks inside the enterprise (intranet) and between an enterprise and its trading partners (extranet) have become the primary information technology infrastructure that supports the e-business applications of many companies. Businesses today depend on the Internet, intranets, and extranets to implement and manage innovative e-business applications. These companies rely on e-business applications to (1) reengineer internal business processes, (2) implement electronic commerce systems with their customers and suppliers, and (3) promote enterprise collaboration among business teams and workgroups.

The most common implementation of e-business is as an additional, or in some cases primary, storefront. By selling products and services online, an e-business is able to reach a much wider consumer base than any traditional brick-and-mortar store could ever hope for.

This function of e-business is referred to as e-commerce, and the terms are occasionally used interchangeably. An e-business may also use the Internet to acquire wholesale products or supplies for in-house production. This facet of e-business is sometimes referred to as e-procurement, and may offer businesses the opportunity to cut their costs dramatically. Even many e-businesses which operate without an electronic storefront now use e-procurement as a way to better track and manage their purchasing.

In addition to buying and selling products, e-business may also handle other traditional business aspects. The use of electronic chat as a form of technical and customer support is an excellent example of this. An e-business which uses chat to supplement its traditional phone support finds a system which saves incredible amounts of time while providing opportunities unavailable through traditional support.

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program, all pertinent information about the hardware and software specifications for a user's computer may be relayed to the support operator directly, without having to walk a customer through personally collecting the data.

Using email and private websites as a method for dispensing internal memos and white sheets is another use of the Internet by e-business. Rather than producing time-intensive and costly physical copies for each employee, a central server or email list can serve as an efficient method for distributing necessary information. In the past few years, virtually all businesses have become, to some degree or another, an e-business. The pervasiveness of Internet technology, readily available solutions, and the repeatedly demonstrated benefits of electronic technology have made e-business the obvious path. This trend continues with new technologies, such as Internet-enabled cell phones and PDAs, and the trend of e-business saturation will most likely continue for some time.

For business, e-commerce includes:

- Performing transactions with customers over the Internet for purposes such as home shopping, banking, and electronic cash use.
- Performing transactions with other organizations through the use of electronic data interchange (EDI)-the direct computer-to-computer transfer of transaction information contained in standard business documents such as invoices and orders.
- Gathering information relating to consumer market research and competitors (called competitive scanning).
- Distributing information to prospective customer through interactive advertising, sales, and marketing efforts.

E-commerce is also giving rise to many best business practices, such as telecommuting and the virtual workplace. Telecommuting and the virtual workplace go hand in hand.

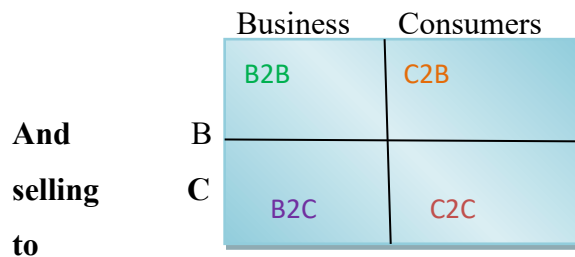
**Telecommuting** is the use of communications technologies (such as the Internet) to work in a place other than a central location.

The **virtual workplace** is a technology enabled workplace. No walls. No boundaries. Work anytime, anyplace, linked other people and information you need, wherever they are.

There are four main perspectives for e-commerce: Business to Business (B2B), Business to Consumer (B2C), Consumer to Business (C2B), and Consumer to Consumer (C2C).

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### Business originating from



**Fig.2.2: Four Categories of E-Commerce**

B2C are the e-commerce sites that sell products and service, or provide information services directly to consumers. They include such well-known sites as Yahoo, Amazon.com, and LandsEnd.com. Even though most of the media's attention focused on B2C, the dollar volume of e-commerce is widely expected to be concentrated in the B2B segment. This is because the B2B segment is so much larger, regardless of whether we're talking about e-commerce or traditional commerce. The C2C sector is where consumers deal with each other through an auction site such as eBay or directly in one of the peer to peer networking applications. Finally, the C2B sector is the one in which the Internet makes it possible for many consumers who want to buy the same or similar products to band together in order to obtain volume discounts from a business. This process is known as demand aggregation. ***Demand aggregation*** combines purchase requests from multiple buyers into a single large order which justifies a discount from the business.